Venkata Lakshmi Parimala Pasupuleti

Senior Full-Stack Engineer · Healthcare AI · Python / FastAPI · React · FHIR · LLM Applications · Async Workflows · AWS · PostgreSQL

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github.com/venkatalakshmiparimala/AgentProbe | venkatalakshmiparimala.github.io/Portfolio | Virginia, USA (Remote)

PROFESSIONAL SUMMARY

Senior full-stack engineer with 8 years shipping production systems and deep alignment with Curai's mission: making quality healthcare accessible through technology. Built **AI on FHIR** — a full-stack platform (React + Python/Flask) integrating LLMs with FHIR healthcare APIs under **HIPAA-compliant RBAC**, reducing manual clinical retrieval time by ~60%. Built **Leasel** — a production Python/FastAPI system with **async agentic workflows**, background job orchestration, idempotency guarantees, RAG retrieval, and **eval-driven iteration** (CI regression harness, LLM-as-judge scoring). Built **AgentProbe** — an open-source **LLM evaluation toolkit** demonstrating eval-driven development applied to non-deterministic systems. At Entrata (8yr): owned **async processing systems** (RabbitMQ background jobs, retry logic, idempotency) serving 100K+ users, **PostgreSQL** schemas for complex eligibility and compliance workflows, **HIPAA-compliant data pipelines** with zero security incidents, and full-stack modules (Python/PHP/Java backend + JavaScript/React frontend) shipped end-to-end across 120+ Agile sprints. Python primary; FastAPI, React, PostgreSQL, DynamoDB, AWS, Docker, Terraform, CI/CD.

CORE COMPETENCIES

Python / FastAPI / Starlette (async) | React (components, hooks, state management) | Async Workflows & Background Jobs | RabbitMQ / Message Queues | Idempotency & Eventual Consistency | PostgreSQL & SQL Optimization | DynamoDB / NoSQL | FHIR / Healthcare Data APIs | HIPAA / PHI Compliance | LLM Applications & Tool Calling | Eval-Driven Iteration | RAG & Retrieval Pipelines | LangGraph / LangChain | REST APIs & API Design | AWS (S3 / Lambda / RDS / SQS / EC2) | Docker / Kubernetes | Terraform | CI/CD (GitHub Actions) | Unit / Integration / E2E Testing | Multi-Tenant Architecture | Java / TypeScript / JavaScript | Agile / Scrum

EXPERIENCE

Blue Horse Digital

Sep 2026 - Present

AI/ML Engineer (Volunteer) — Virginia, USA

- Building AI-enabled applications and data pipelines; integrating LLM components with backend and React frontend systems; surfacing insights into application-facing APIs.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an agentic workflow system (LangGraph stateful graph) with RAG over a regulatory corpus, validation guardrails, and MCP server deployment - delivering structured, cited outputs into downstream tooling via async pipeline.

Entrata — PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Owned async processing systems end-to-end**: RabbitMQ-based background job pipeline processing gross-rent updates across thousands of tenant records — designed **retry logic, idempotency guarantees, and dead-letter handling** that eliminated double-processing failures; maintained through production incidents and annual rent-schedule changes. The same async reliability patterns a healthcare messaging platform requires.
- **Full-stack module delivery across 120+ Agile sprints**: Python/PHP/Java backend services + JavaScript/React frontend components shipped end-to-end, owning features from design through delivery and continued iteration; 20+ production modules shipped with unit, integration, and regression coverage.
- **HIPAA-compliant data pipelines**: 100% PHI compliance, Libsodium-encrypted pipelines, multi-tenant data isolation at the schema layer; zero security incidents across full AWS S3 migration serving 100K+ monthly active users.
- Optimized 200+ PostgreSQL queries (30% performance gain) across complex multi-tenant eligibility and compliance schemas; mentored 5+ engineers; 300+ code reviews; 80% reduction in manual operational workload via automation pipelines.

Entrata India — Db Xento Systems

Jan 2017 - Sep 2018

Associate / Trainee Software Engineer — Pune, India

- Built compliance and document management modules (PHP, Python, Java, PostgreSQL); led AWS S3 cloud migration; income-limits import (1.6M records across 5 years).

PROJECTS

AI on FHIR: Full-Stack Healthcare Data Integration Platform

FHIR · React · Python · LLMs · HIPAA · Healthcare AI

Full-stack platform (React frontend + Python/Flask backend) integrating **LLMs with FHIR healthcare APIs**: translated natural-language queries (EN/ES) into structured FHIR API calls using LLMs, BERT, and embeddings; enforced **HIPAA-compliant RBAC** across structured medical records. **~60% reduction in manual clinical retrieval time**. Directly relevant to Curai: LLM-powered interface over structured healthcare data, bilingual, access-controlled, compliance-first.

Python · Flask · React · LLMs · BERT · FHIR APIs · Docker · spaCy · 2024

Leasel: Production AI Platform — Async Workflows & Eval-Driven Iteration

FastAPI · LangGraph · Async · RAG · LLM Eval · CI/CD

Production Python/FastAPI agentic system with **async orchestration**, background processing, and **eval-driven iteration**: 8-agent LangGraph StateGraph, tool calling, hybrid RAG (TF-IDF + semantic reranking), citation grounding, and conditional HITL routing. **Evaluation harness**: 5-metric engine (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, CI-integrated regression gate — every prompt or pipeline change runs the full eval suite before merge. 93.6% precision on 1,240 test cases. Built and iterated using eval results, not intuition.

Python · FastAPI · LangGraph · ChromaDB · OCR · SQLite · GitHub Actions · 2026

AgentProbe: Open-Source LLM Evaluation Toolkit

Eval-Driven · Multi-Provider · CI/CD · Hallucination Detection

Reusable toolkit for **eval-driven LLM development**: 5-metric eval engine, LLM-as-judge scoring, CI regression harness with golden datasets, offline + online evaluation loops, and multi-provider routing (OpenAI, Anthropic, Groq, Ollama). Ships the discipline of defining evaluation metrics before writing the first prompt — applicable directly to LLM-powered clinical features that must be observable, measurable, and safe to ship multiple times daily. Live: github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · 2025

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia May 2026 I GPA 3.87

B.Tech Computer Science & Engineering — JNTU, Anantapur 2017